

Dear Editor,

After a careful review of the manuscript “**Evaluating the impact of a synthetic predator pheromone, a novel pest management tool, on target pests and insect communities in potato fields**” I feel that this manuscript, as it stands with major revisions, can be acceptable for publication in the Journal of Insect Science.

The study addresses an innovative application of predator-derived semiochemicals and combines field and greenhouse experiments to evaluate both ecological and applied consequences. The results suggest that the synthetic *Podisus maculiventris* pheromone can enhance biological control of *Leptinotarsa decemlineata* without major community-level disruptions, which is an important contribution.

The introduction provides a fair, generalized background of the topic that gives the reader an appreciation of the full range of applications for this experiment. The title and abstract are appropriate for the content of the text. However, I left comments with line numbers below, that need to be addressed.

The methodology used in this study was sound. I, however, had several comments that need attention. I specifically wonder about the field design and statistical unit, I believe it needs clearer justification. For instance, sentinel egg predation analyses may suffer from pseudoreplication. In addition, community analyses are conducted at family level, which may obscure species-level responses. Several methodological details are insufficiently described.

The result section was clear and easy to follow, with enough detail to understand the findings of this study.

The discussion is well organized and effectively relates the findings to previous research on predator-derived semiochemicals and biological control. However, some conclusions regarding the mechanism of increased egg predation are not directly supported by the data and seems to be a stretch. This is mainly because predator abundance did not differ between treatments and predator activity was not directly measured in the field, the suggestion that enhanced biological control resulted primarily from increased predator foraging behavior should be discussed cautiously, as its mainly speculative without actual data. The authors should also briefly acknowledge limitations of the study, including its single location and season, and discuss how these may affect the broader applicability of the findings.

Overall, the Manuscript is well-written, the literature is well-reviewed, and the entire manuscript is clearly structured. I enjoyed reading this paper, and I believe that this manuscript, with major edits, could be suitable for publication in the Journal of Insect Science.

Regards,

Specific Comments with line numbers

Abstract

Lines 17–20

The statement that predation rates were "five times higher" should include actual means or percentages to provide biological context.

Lines 18–20

The abstract implies that enhanced predation explains beetle suppression. This conclusion is plausible but not directly demonstrated because larval abundance and egg predation were measured independently.

Introduction

Lines 34–40

Consider adding another example of successful aggregation pheromone use in open-field agriculture beyond cucumber beetles and thrips.

Lines 58–65

The discussion of non-consumptive effects would benefit from mentioning that predator cues can also induce physiological responses in plants, which may indirectly affect herbivores.

Lines 94–103

The hypotheses are not clearly stated. Consider explicitly listing predictions, For instance: reduced CPB abundance, altered herbivore community composition, altered predator abundance or diversity, increased predation rates.

Lines 97–98

The sentence appears incomplete:

"pheromone-induced changes in the abundance or feeding behavior of naturally occurring predators."

Please revise for clarity.

Methods

Field Design

Lines 128–133

Please provide a stronger justification for the selected release rate and explain how it compares to natural pheromone emissions under field conditions. The authors mentioned something about 1 mg is likely what CPB experience in the field, but that assumption is not entirely valid, and there will likely be multiple female that can simultaneously release the pheromones.

Lines 137–138

Five-meter buffers may not fully prevent pheromone contamination between plots. Please discuss why this distance was considered sufficient.

Field Sampling

Lines 145–158

Were the same observers used throughout the study? If multiple observers participated, explain how observer bias was handled.

Lines 147-148

What do the authors mean by all CPB predators? Do you have a knowledge of what species feed on CPB, eggs and larvae? This needs to be better explained here, as such information requires molecular gut content analysis to count a species as predator of CPB. Anyhow, detail information regarding what species were considered CPB predators is needed, to better understand and evaluate this method.

Major concern: Carabid beetles are among the most important predatory groups of CPB, and none of the proposed predatory data collection techniques will reliably estimate their density in potato plots. These predators are known to be nocturnal, and hence pitfall traps are the only reliable method for their collection. How do authors justify the techniques used in this experiment? Why did not include pitfalls?

Lines 153–154

Since insects were not collected, describe how field identifications were validated. How skilled the team were, and how that was factored into data accuracy.

Insect Identification

Lines 160–175

Family-level identification may mask treatment effects on particular predator species. Please justify why family-level resolution was sufficient for community analyses.

Lines 173–175

Excluding taxa representing less than 1% of observations could bias richness and diversity estimates. Please explain how this exclusion may influence results. Given the sight identification in fields, how confident were the team on accurately selecting these taxa?

Sentinel Egg Predation**Lines 183–191**

Were sentinel plants randomized among plots each week? Also, authors should clarify whether repeated deployments occurred in the same locations.

Lines 184–185

The sample size description is confusing. Please specify: 1) number of plots sampled per week, 2) number of egg clutches per treatment per week, and 3) total number of observations included in analyses.

Lines 189–191

This adjustment for background mortality is appropriate, but it would be useful to report average non-predation egg mortality, to achieve a more realistic number attributed to predation.

Greenhouse Assays**Lines 213–214**

Relative humidity (21.4%) appears unusually low for greenhouse bioassays involving predators and eggs. Please verify this value, how did the authors achieve such low humidity in the greenhouse.

Lines 221–223

Include trial identity as a random effect if experiments were conducted sequentially.

Lines 225–229

Sample sizes are highly unbalanced among predator species. Discuss whether statistical power differed among taxa. Also, provide justification as to why the sample size effect is not major issue in this study.

Statistical Analyses

Lines 236–238

Clarify whether week was treated as a random or repeated factor. Temporal autocorrelation should be considered because plots were sampled repeatedly. Repeated measures are a standard way of analyzing this.

Lines 245–247

Community metrics were calculated using family-level taxa. Consider discussing limitations associated with this approach.

Lines 277–280

The egg predation model appears to treat observations as independent. Because plots were sampled repeatedly through time, plot should likely be included as a random effect, as they are not true independent observations.

Results

Lines 299–302

Please report effect size estimates (e.g., percentage reduction) in addition to statistical significance.

Lines 322–323

The claim of increased predation is compelling. Consider reporting estimated marginal means and confidence intervals.

Lines 325–330

Report actual egg consumption percentages along with your statistical test results.

Discussion

Lines 332–343

The discussion attributes increased field predation primarily to altered predator behavior.

Since predator activity was not directly measured, this conclusion should be presented more cautiously.

Lines 362–372

An important alternative explanation is that D-vac and visual sampling may not adequately detect localized changes in predator activity. So, it is always wise to discuss the study limitations.

Lines 375–380

The proposed explanation involving increased feeding under perceived competition is interesting but speculative. Consider emphasizing that this mechanism remains untested or present it as a future direction.

Lines 397–410

The applied implications are well presented, but the discussion would benefit from acknowledging limitations, including only one year of data, one location, one cropping system and one pheromone applied rate (which, I had an earlier comment about how the authors selected this rate).

Figures and Tables

Figure 1

Include sample size and actual mean $\pm$ SE values in either figure or caption.

Figure 3

Correct the discrepancy between the caption and text regarding predator groups shown.

Figure 4

Report exact means and confidence intervals.

Figure 5

Consider displaying raw data points or boxplots rather than means $\pm$ SE to illustrate variation.

Minor Editorial Corrections

Line 75

Add a space before citation:

"(Leptinotarsa decemlineata) (Say)"

Line 340

Remove extra space before comma: "adults , lacewing larvae"

Line 367 Replace: "HIPV's" with "HIPVs"

Line 505: Verify citation year for *emmeans* package (2025).